

Influence of Future Predictions over Active Learning of Users' Tastes for Recommender Systems

Task 1: Guided Active Learning for Cold-Start Recommendation

Objective

To design a recommender system that helps new users build their profile interactively, while seeing the effect of their ratings on future recommendations.

Dataset

We use the **MovieLens small latest 100K** dataset, which includes 100,000 ratings across 9,000 movies from 600 users.

Recommendation Model: Matrix Factorization

We use matrix factorization to model user-item interactions:

$$\min_{U,V} \sum_{(i,j) \in K} (R_{ij} - U_i^T V_j)^2 + \lambda(\|U\|_F^2 + \|V\|_F^2)$$

- U_i : latent vector for user i
- V_j : latent vector for movie j
- λ : regularization hyperparameter

Cold Start User Embedding

I trained the model beforehand using the MovieLens dataset and computed the item embedding matrix V . This matrix was saved to a `.pkl` file.

In the application, I load this saved `.pkl` file containing the movie embeddings V . Then, based on the new user's submitted ratings, I compute the user embedding vector U_{new} by solving the following optimization problem:

$$\min_{U_{new}} \sum_{j \in K_i} (R_{ij} - U_{new}^T V_j)^2 + \lambda \|U_{new}\|_F^2$$

This allows the recommender system to quickly adapt to the new user's preferences without retraining the entire model.

Guided Active Learning Strategy

1. Initial Movie Selection (Diversity Sampling):

For the first 5 ratings, movies are selected to maximize diversity. Specifically, the system shows movies whose embeddings are most dissimilar (lowest cosine similarity) to the average embedding of the movies the user has already rated, ensuring broad genre and taste coverage.

2. After 5 Ratings - Active Learning (Uncertainty Sampling):

Once the user has rated at least 5 movies, the next movie is selected based on uncertainty sampling. The system predicts the user's rating for unrated movies using the current user embedding and movie embeddings, then selects the movie with a predicted rating closest to the neutral score of 3.0. This targets movies for which the system is most uncertain about the user's preference, to better refine the user model.

3. User Input:

- The user rates each movie from 0.5 to 5 stars or can choose to skip rating up to 5 times.
- After 5 skips, the skip option is disabled to encourage participation.
- If the user attempts to submit without selecting a rating, an alert prompts them to enter a rating first

Real-Time Visualization

After each user rating, the system updates two visualizations that reflect the user's evolving preferences:

• Bar Chart - Average Rating per Genre:

For each genre, the chart shows the average of the **rating values** the user has given to movies in that genre.

This is computed as:

$$\text{Average Genre Rating}_g = \frac{\sum \text{Ratings Given to Movies in Genre } g}{\text{Number of Ratings in Genre } g}$$

This helps identify how positively the user rates different genres.

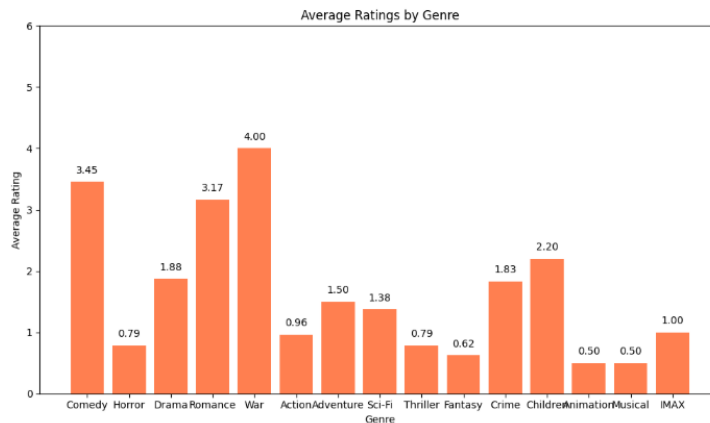


Figure 1: Bar Chart displaying average user ratings across different movie genres

- **Radar/Spider Chart - Genre Influence Profile:**

This chart visualizes how much each genre contributes to the user’s overall taste profile. It is computed as:

$$\text{Genre Influence}_g = \frac{\sum \text{Ratings Given to Movies to Genre } g}{\text{Total Number of Ratings Submitted by User}}$$

This normalization reflects the user’s engagement or preference toward particular genres - a higher value means more attention or liking toward that genre.

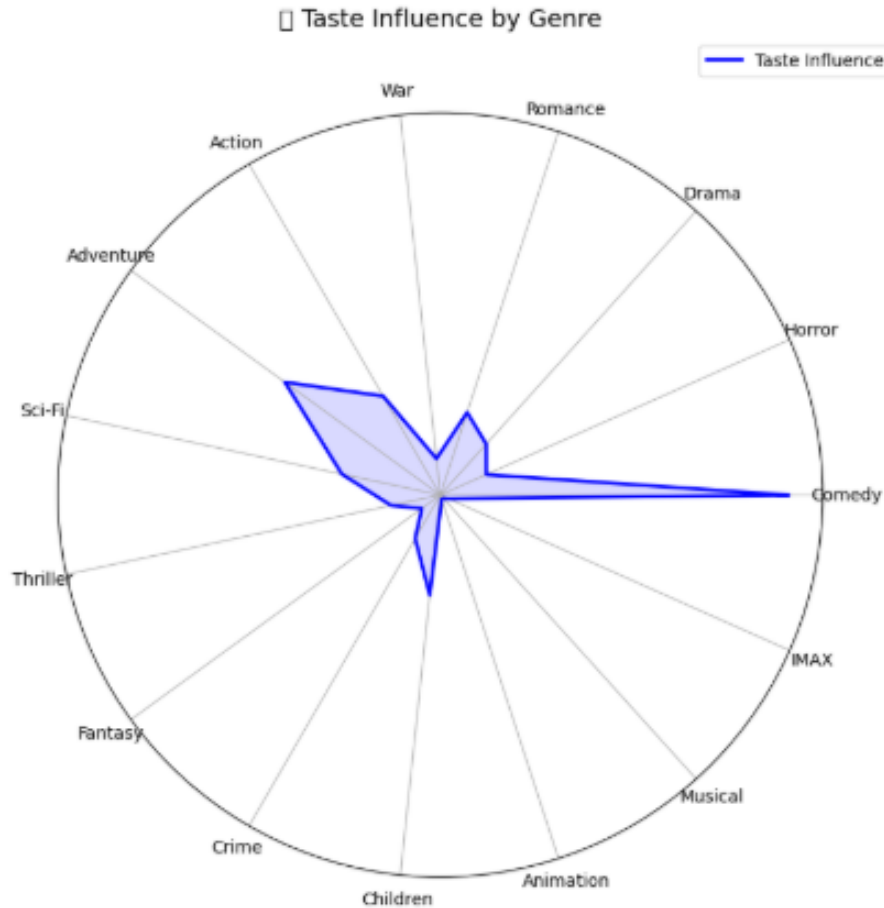


Figure 2: Radar Chart illustrating the influence of each genre on the user’s taste profile

These visualizations are updated in real-time and help guide the user during the interactive preference-building process.

Influence-Based Recommendation Preview

Before rating a movie, the user is shown a preview message such as:

“If you rate this movie as 5 stars, you may also like: Movie A, Movie B, Movie C.”

This feature demonstrates how the current rating could affect future recommendations. It encourages the user to think strategically while rating, enhancing engagement and transparency in the recommendation process.

How it works:

The system simulates the effect of a 5-star rating for the current movie by temporarily updating the user embedding with that hypothetical rating. Based on this updated embedding, the model generates the top 3 recommendations as a preview.

Handling Rating Bias (Strict vs. Generous Users)

Different users have different rating habits:

- Some users are generous and tend to give high ratings, usually between 3 to 5.
- Others are stricter and rarely give ratings above 4.

These personal biases can skew the model's understanding of user preferences.

To normalize for such tendencies, we subtract the **global mean rating** (average rating across all users and movies) from each user's rating before training or prediction. This centers the ratings around zero and helps ensure that the recommender system focuses on **relative preferences** rather than absolute scores.

This adjustment improves fairness and consistency in modeling user tastes.

Task 2: Conceptual User Study Design

Hypothesis

Providing users with feedback about how their ratings influence recommendations increases both rating engagement and profile accuracy.

Study Design (Conceptual Only – Not Executed)

We conceptualize a **between-subjects study** with two groups:

- **Control Group:** Users rate movies without receiving any feedback about how their ratings influence recommendations.
- **Experimental Group:** Users receive real-time influence previews (e.g., "If you rate this movie 5 stars, you may also like...") before rating each movie.

Each participant would interact with the system until they have submitted at least **10 ratings**.

Recruitment Strategy

Participants would be recruited from the following sources:

- **University mailing lists** to target a broad student population to ensure varied backgrounds
- **Online forums and social media platforms** related to movies, technology, or recommendation systems (e.g., Reddit, Discord, LinkedIn groups)

This strategy aims to include users with diverse perspectives, from casual movie watchers to those familiar with AI or recommender systems.

Theoretical Data Points to Collect

If the study were conducted, the following metrics would be tracked:

- Number of ratings submitted
- Number of skips used
- Genre diversity in ratings
- Time taken per rating

Note: No actual data is collected, as this is a conceptual study.

Task 3: Interactive System Implementation

Interface Summary

- Landing Page
 - Button to download PDF overview
 - Button to enter the recommender system
- Recommender System
 - Movie display (poster, title, backdrop, overview)
 - Star-based rating system (supports half-star precision)
 - Skip button (up to 5 total skips allowed)
 - Alert if no rating is selected
 - Real-time updated bar chart and spider (radar) chart
 - Future influence preview:
 - * “If you rate this 5 stars, you may also like: Movie A, Movie B, Movie C”
 - Final top-10 movie recommendations displayed **after the user submits at least 10 ratings**

Summary

This project presents a user-guided cold-start recommender system that integrates active learning and future impact visualization to enhance recommendation accuracy and user engagement. Through the design of a conceptual user study and the development of an interactive system, this report proposes a novel, human-centric approach to addressing the cold-start problem in recommender systems.